

DATA SCIENCE — DETAILED MASTER INDEX

Detailed Master Index • B.Tech CSE 4th Year • Placement / Interview / OA Preparation

0. COMMON CSE + INTERVIEW FOUNDATION

- 0.1 Programming [B][INT][OA]** — Variables, data types, operators, input/output, conditionals, loops, functions, scope, recursion, exceptions and debugging.
- 0.2 Python [B][INT][OA]** — Lists, tuples, sets, dictionaries, strings, slicing, comprehensions, lambda, map/filter/reduce, iterators, generators and modules.
- 0.3 OOP [B][INT]** — Class/object, constructor, encapsulation, inheritance, polymorphism, abstraction, composition, overriding, overloading concepts and SOLID basics.
- 0.4 DSA [B→A][INT][OA]** — Arrays, strings, hashing, linked lists, stacks, queues, trees, BST, heap, trie, graphs, recursion, backtracking and dynamic programming.
- 0.5 Algorithms [B→A][INT][OA]** — Sorting, binary search, two pointers, sliding window, prefix sums, greedy, divide-and-conquer, BFS/DFS and DP.
- 0.6 Complexity [B][INT][OA]** — Big-O, Big-Theta, Big-Omega, time/space complexity, recursion complexity and amortized analysis.
- 0.7 SQL [B→A][INT][OA]** — SELECT, WHERE, GROUP BY, HAVING, joins, subqueries, CTEs, CASE, windows, ranking, dates, NULLs and query optimization.
- 0.8 Git/Linux [B][P]** — Git workflow, branches, merge/rebase, GitHub, CLI, permissions, processes, pipes, grep/sed/awk and environment variables.
- 0.9 Interview method [INT]** — Clarify → approach → brute force → optimize → code → test edge cases → complexity → alternatives → trade-offs.

1. DATA SCIENCE FOUNDATIONS [B][INT]

- 1.1 Data Science definition** — Statistics + programming + domain knowledge + communication for data-driven decisions.
- 1.2 Lifecycle** — Business problem → data acquisition → cleaning → EDA → modeling → evaluation → communication → deployment.
- 1.3 Problem framing** — Stakeholders, objective, KPI, constraints, assumptions and success criteria.
- 1.4 Data types** — Numerical, categorical, ordinal, temporal, text, image and semi-structured data.
- 1.5 Data quality** — Accuracy, completeness, consistency, uniqueness, validity and timeliness.

2. PYTHON + DATA LIBRARIES [B→A][OA]

- 2.1 Python core** — Syntax, collections, functions, comprehensions, exceptions, modules and OOP essentials.
- 2.2 NumPy** — Arrays, dimensions, indexing, broadcasting, vectorization and linear algebra operations.
- 2.3 pandas** — Series/DataFrame, filtering, indexing, groupby, merge, concat, pivot and reshaping.
- 2.4 Cleaning** — Missing values, duplicates, types, string/date cleaning and transformations.
- 2.5 Visualization** — Matplotlib/Seaborn concepts, plot types, annotations and chart selection.
- 2.6 Notebooks/environments** — Jupyter, virtual environments, package management and reproducibility.

3. SQL [B→A][INT][OA]

- 3.1 Fundamentals** — SELECT, WHERE, ORDER BY, LIMIT, DISTINCT, expressions and NULL behavior.
- 3.2 Aggregation** — COUNT, SUM, AVG, MIN/MAX, GROUP BY and HAVING.
- 3.3 Joins** — INNER, LEFT, RIGHT, FULL, CROSS and self joins.
- 3.4 Subqueries/CTEs** — Nested queries, correlated queries and WITH clauses.
- 3.5 Windows** — ROW_NUMBER, RANK, DENSE_RANK, LAG, LEAD, running totals and partitions.
- 3.6 Advanced SQL** — CASE, date/string functions, set operations and query debugging.
- 3.7 Performance** — Indexes, execution plans and avoiding expensive queries.

4. STATISTICS + PROBABILITY [B→A][M][INT][OA]

4.1 Descriptive statistics — Mean, median, mode, variance, standard deviation, percentiles and IQR.

4.2 Probability — Events, conditional probability, independence and Bayes theorem.

4.3 Distributions — Normal, binomial, Bernoulli, Poisson, uniform and exponential.

4.4 Sampling — Population, sample, sampling methods, bias and central-limit intuition.

4.5 Estimation — Point estimates, standard error and confidence intervals.

4.6 Hypothesis testing — Null/alternative, p-value, significance, Type I/II errors and power.

4.7 Correlation — Pearson, Spearman, covariance and correlation-vs-causation.

4.8 A/B testing — Experiment design, randomization, metrics, significance and practical impact.

5. EDA [B→I][INT][P]

5.1 Univariate — Distributions, frequency, spread and outliers.

5.2 Bivariate — Numeric-numeric, categorical-numeric and categorical-categorical relationships.

5.3 Multivariate — Interactions, correlation matrices and high-dimensional patterns.

5.4 Missingness — Missing patterns, mechanisms and appropriate treatment.

5.5 Outliers — Detection, visualization, investigation and domain validation.

5.6 EDA storytelling — Observation → hypothesis → analysis → insight → action.

6. MACHINE LEARNING [B→A][INT][OA]

6.1 Supervised learning — Regression/classification framing, baselines and data splits.

6.2 Regression — Linear, Ridge, Lasso and tree-based approaches.

6.3 Classification — Logistic regression, k-NN, Naive Bayes, trees, random forest and SVM.

6.4 Ensembles — Bagging, boosting, gradient boosting and XGBoost concepts.

6.5 Unsupervised — K-means, hierarchical clustering, DBSCAN and PCA.

6.6 Preprocessing — Encoding, scaling, feature engineering, selection and pipelines.

6.7 Evaluation — Regression/classification metrics, CV, leakage, overfitting and model selection.

6.8 Explainability — Feature importance, permutation importance and SHAP/LIME concepts.

7. ADVANCED DATA SCIENCE [I→A][INT]

7.1 Time series — Trend, seasonality, stationarity, lag/rolling features and forecasting concepts.

7.2 NLP — Text cleaning, tokenization, n-grams, TF-IDF, embeddings and text classification.

7.3 Recommenders — Content-based, collaborative filtering, matrix factorization and ranking concepts.

7.4 Causal thinking — Confounding, selection bias, interventions and causal-vs-correlational reasoning.

7.5 Experimentation — A/B tests, metrics, guardrails, novelty effects and experiment pitfalls.

7.6 Data engineering — ETL/ELT, batch/streaming, warehouses, lakes and pipelines.

7.7 Big data — Distributed processing, Spark concepts, partitions, shuffles and joins.

7.8 Cloud data — Object storage, compute, managed databases and analytics services.

8. VISUALIZATION + BUSINESS COMMUNICATION [INT][P]

8.1 Chart selection — Match charts to comparison, distribution, relationship, composition and trend.

8.2 Dashboards — KPIs, filters, drill-downs, layout and stakeholder requirements.

8.3 Storytelling — Context → evidence → insight → recommendation → expected impact.

8.4 Visualization mistakes — Misleading axes, clutter, bad scales, inappropriate charts and overclaiming.

8.5 Presentation — Explain technical findings clearly to technical and non-technical audiences.

9. PROJECTS + INTERVIEW/OA [P][INT][OA]

9.1 EDA project — Messy dataset → cleaning → statistical analysis → visualization → insights.

9.2 Predictive project — Business problem → SQL/data → EDA → ML → evaluation → recommendation.

9.3 Dashboard project — Data source → transformation → KPI design → dashboard → business story.

9.4 End-to-end project — Ingestion → processing → warehouse → model → API/dashboard → documentation.

9.5 Case interviews — Product metrics, funnels, retention, conversion, cohorts and experiment design.

9.6 Technical interviews — Python, SQL, statistics, ML, EDA and project defense.

9.7 OA preparation — DSA + Python + SQL + statistics + ML MCQs + timed case questions.

9.8 Behavioral — Ownership, failure, conflict, learning, teamwork and communication.

Final preparation sequence: Fundamentals → Core theory → Implementation → Problem solving → Projects → Deployment → Technical interview → OA → Mock interview → Revision.